Manish Karki

DevSecOps Engineer

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SUMMARY

DevSecOps Engineer who runs and secures bare-metal, fintech-grade production infrastructure: self-hosted Kubernetes, Jenkins CI/CD, HAProxy and Nginx load balancing, PostgreSQL and Zero Trust controls. Came up through offensive security (penetration testing, vulnerability assessment, red-team simulation), so systems get built the way an attacker would test them. Automates with Go, Python and Bash; creator of InfraEye, an open-source observability platform.

SKILLS

DevOps & Platform: Kubernetes (bare metal), Docker, BuildKit, Helm, Jenkins, GitHub Actions, CI/CD, HAProxy, Nginx, Infrastructure as Code, Linux administration
Security: Penetration Testing, Vulnerability Assessment, Zero Trust, IAM, OWASP Top 10, MITRE ATT&CK, CVE analysis, Wazuh (SIEM), Nessus, Burp Suite, Nmap, Metasploit, Wireshark
Cloud & Data: AWS (security, architecting, operations), PostgreSQL, Redis, SQLite
Languages: Go, Python, Bash, TypeScript, React, SQL, Java, C++
Networking: TCP/IP, DNS, VPN, OSPF, firewalls (nftables), packet analysis

EXPERIENCE

DevSecOps Engineer, Citytech | Kathmandu, Nepal | May 2026 – Present

- Operate and harden bare-metal infrastructure for a banking/fintech production system, owning provisioning, hardening and lifecycle management without managed cloud services.
- Deploy and run microservices on self-hosted Kubernetes; build and maintain Jenkins CI/CD pipelines for build, test and release.
- Eliminated recurring deployment downtime on the bank's live services by designing a rolling-deployment strategy behind HAProxy and Nginx with health-check-gated cutover, achieving near-zero-downtime releases.
- Administer PostgreSQL behind the microservices, keeping data available through every deployment cycle.
- Introduced Zero Trust controls: least-privilege access, network segmentation and service-level authentication.

Offensive Security Intern, InRED Labs | Kathmandu, Nepal | Jul 2025 – Oct 2025

- Performed vulnerability assessments and penetration tests on internal web applications and network infrastructure, from recon and enumeration through exploitation and post-exploitation (Nmap, Burp Suite, Metasploit).
- Reported findings with CVE references and remediation guidance; automated recon and reporting in Python and Bash.
- Ran realistic attack simulations with the blue team, mapped to MITRE ATT&CK, including privilege-escalation testing.

IT Support and SOP Reviewer, Uni Express | Kathmandu, Nepal | Feb 2023 – Dec 2024

- Managed IT infrastructure and endpoint protection; reviewed 100+ academic Statements of Purpose.

IT Technician and SOP Reviewer, Yes International Education | Kathmandu, Nepal | Nov 2018 – Feb 2021

- Maintained systems and network infrastructure, implemented security protocols, and ran backup and recovery.

PROJECTS

InfraEye | Go, React, TypeScript, PostgreSQL, Redis, Wails | github.com/mnshchtri/infra-eye

- Agentless infrastructure observability platform (creator and maintainer): real-time telemetry across bare metal and Kubernetes, and a self-healing engine that turns alert rules into automated SSH remediation.
- Kubernetes diagnostics, AI-driven log analysis over LLM/MCP, OIDC single sign-on (Keycloak, Auth0, Okta, Azure AD), and a native macOS/Linux desktop app shipped via Homebrew.

DeployForge | Kubernetes, BuildKit | private

- Self-hosted deployment platform: builds each commit with BuildKit, rolls out beside the live version and switches traffic only once healthy; preview deployments, one-click rollback, encrypted secrets, custom domains.

TrimurtiSec | Python: penetration-testing framework with recon, exploitation and persistence modules and CVE-mapped reports.
secaudit | Bash: read-only security audit for Arch Linux that scores the machine and suggests safe fixes.
Wazuh SIEM: deployed agents and centralised logging; wrote and tested alert and correlation rules across hosts.

EDUCATION

BSc (Hons) Computer Networking and IT Security, Islington College, London Metropolitan University (completed)
B.E. Electronics and Communication Engineering, Sagarmatha Engineering College, IOE (discontinued)

CERTIFICATIONS

AWS Academy: Cloud Security Foundations, Cloud Architecting, Cloud Operations, Cloud Foundations
TCM Security: Practical Ethical Hacking | TryHackMe: SDLC Security, Docker Security, Pre-Security
30+ TryHackMe and Hack The Box labs; regular CTFs (web, privilege escalation, scripting)